

House Price Prediction using Linear Regression

1. Introduction

This project aims to build and evaluate a Linear Regression model using the California Housing Dataset. The objective is to predict median house values based on various housing-related features and understand the complete machine learning workflow, including data exploration, model training, evaluation, and result analysis.

2. Dataset Description

The California Housing Dataset was obtained from Scikit-Learn. It contains information collected from California districts.

Dataset Information

- Total Records: 20,640
- Features: 8
- Target Variable: MedHouseVal (Median House Value)

Features Used

1. MedInc – Median Income
 2. HouseAge – Median House Age
 3. AveRooms – Average Rooms
 4. AveBedrms – Average Bedrooms
 5. Population – Population
 6. AveOccup – Average Occupancy
 7. Latitude
 8. Longitude
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3. Exploratory Data Analysis (EDA)

Several exploratory data analysis techniques were performed to understand the dataset.

Findings

- No missing values were found in the dataset.
- House prices showed a slightly skewed distribution.
- Correlation analysis indicated that Median Income (MedInc) has the strongest positive relationship with house prices.
- Scatter plots showed that higher-income regions generally have higher housing prices.

Visualizations Used

- House Price Distribution Histogram
 - Correlation Heatmap
 - Income vs House Price Scatter Plot
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4. Model Development

Algorithm Used

Linear Regression

Data Splitting

- Training Data: 80%
- Testing Data: 20%

The model was trained using the training dataset and evaluated on unseen testing data.

5. Model Evaluation

The performance of the model was evaluated using the following metrics:

Metric	Value
MAE	(0.533)
RMSE	(0.746)
R ² Score	(0.576)

Interpretation

- MAE measures the average prediction error.
- RMSE penalizes larger prediction errors.
- R² Score indicates how well the model explains the variance in house prices.

The obtained results indicate that the model provides reasonable predictions for housing prices.

6. Results and Discussion

The Actual vs Predicted plot showed that many predictions were close to the ideal prediction line. Residual analysis indicated that errors were distributed around zero, suggesting that the model captures the general trend of the data.

Among all features, Median Income was found to be the most influential factor affecting house prices.

7. Future Improvements

The performance of the model can be improved by:

- Applying feature engineering techniques.
 - Removing outliers.
 - Performing feature scaling.
 - Using advanced regression models such as:
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - XGBoost Regressor
 - Hyperparameter tuning.
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8. Conclusion

This project successfully demonstrated the complete machine learning workflow using the California Housing Dataset. A Linear Regression model was trained and evaluated using standard metrics such as MAE, RMSE, and R² Score. The analysis revealed that Median Income has the strongest impact on house prices. Overall, the model achieved satisfactory performance and provides a strong foundation for exploring more advanced machine learning algorithms in future work.







